

# Synergistically Improved Antifouling Efficiency of a Bioinspired Self-renewing Interface via a Borneol/ Boron Acrylate Polymer

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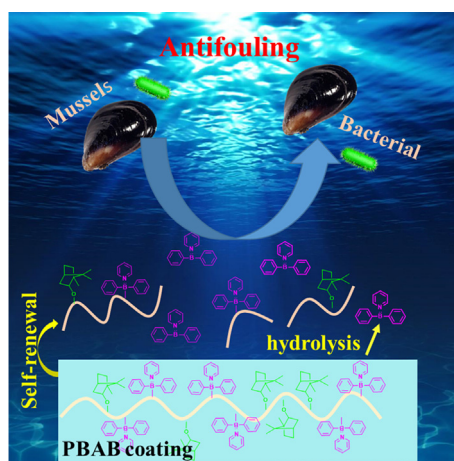
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## HIGHLIGHTS

- Natural borneol groups were introduced into the coating to improve antibacterial properties.
- The addition of boron side chains enables the surface of antifouling coatings to be dynamically updated.
- PBAB antifouling coating can maintain excellent antifouling performance in six months of marine experiments.

## GRAPHICAL ABSTRACT



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## ABSTRACT

Underwater facilities are often perplexed by severe and ubiquitous biofouling. The widely applied commercial antifouling materials still have several challenges in static applications. Herein, a polymer containing isoborneol and borane (PBABs), the borneol derivative structure and grafted pyridine-triphenylborane (PTPB) as antifouling groups were prepared by radical polymerization. PBABs showed high antibacterial rates for *Escherichia coli* (*E. coli*) and *Staphylococcus aureus* (*S. aureus*) of up to 95.1% and 81.1%, respectively, confirming superior antibacterial adhesion properties. More importantly, PBABs effectively reduced the expression of mussel adhesion protein, indicating superior antifouling properties, resulting from the synergistic effect of multiple antifouling functional groups on the material's surface. Therefore, the PBABs have been evaluated as noncytotoxic, low-cost, easily synthesized, and mass-produced, which demonstrates their great potential for actual marine applications.

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## 1. Introduction

Marine biological fouling refers to the attachment of organisms to large or other artificial underwater surfaces [1–2]. Marine

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biological fouling can cause many hazards to marine equipment, including reducing ship speed, accelerating corrosion, shortening ship life, and blocking submarine pipelines, which eventually leads to a waste of resources and huge economic losses[3–7]. Therefore, antifouling coatings are applied on the surfaces of ships and offshore platforms to reduce marine biofouling[8–11]. Traditional antifouling coatings have caused great harm to the environment, and an organotin antifouling coating that has been widely used and effectively reduces biofouling adhesions was explicitly prohibited by the International Maritime Organization (IMO) in 2008 due to serious damage to the marine environment[12–13]. Subsequently, copper-containing antifouling paints replaced organotin coatings but were also restricted by the excessive copper content in ports and offshore[14–18]. Hence, according to the current situation and the restriction of toxic antifouling coatings, it is particularly important to develop environmentally friendly antifouling coatings for marine applications.

Natural biological antifouling agents have become a research hotspot in recent years. Capsaicine[19], indole and its derivatives [20], tannic acid[21], etc. have been gradually applied to marine antifouling coatings. Borneol is a natural medicine that exists in many plants and has been used in antibacterial and marine antifouling fields[22–24]. Borneol has a complex molecular structure, making it an ideal unit of hydrophobic molecular structure, which has a positive impact on its antifouling performance. Luo et al. designed a new type of antibacterial material based on natural borneol molecules, which showed excellent antibacterial properties and good safety[25]. Importantly, borneol molecules grafted onto long-chain molecules do not affect their antifouling activity [26–27]. However, adding borneol and its derivatives directly to the coating faces the problem of an uncontrollable release rate, which reduces the antifouling performance of the coating. Therefore, it is a new challenge to find a suitable method to introduce borneol-based compounds into the resin.

Self-polishing antifouling coatings are a widely used antifouling coating at present[28–31]. Such an antifouling coating immersed in seawater will gradually update the surface, so that the fouling organisms and solid substances attached to the surface will peel off to achieve antifouling[32]. When the coating surface is updated, the surface roughness decreases, and it also plays a certain drag reduction role. In our previous research[33], we synthesized a self-polishing acrylic resin containing boron in the side chain and demonstrated its stable polishing and excellent antifouling properties.

Herein, identical to the traditional method of free radical polymerization, the borneol derivative structure was grafted in the side chain of an acrylic resin, and then pyridine-triphenylborane (PTPB) was added to obtain a polymer containing isoborneol and borane (PBABs). By adopting this tactic, excellent antifouling performance was achieved by adjusting the content of acrylic acid (AA). To evaluate the antifouling performance of the PBAB antifouling coating, static water contact angle, antibacterial performance, immersion weight loss, algae adhesion, and growth rate tests were carried out. The results showed that PBABs have a strong antifouling ability and have little impact on the environment. This work provides a promising strategy for the synthesis and practical application of antifouling coatings.

## 2. Experimental section

### 2.1. Materials

Isobornyl methacrylate (IBOMA, AR), benzoyl peroxide (BPO, AR), agar powder (BR), and 2,4-diphenyl-4-methyl-1-pentene (MSD, AR) were purchased from Aladdin Company. PTPB (indus-

trial grade purity) was purchased from Changzhou Kean Chemical Technology Co., Ltd. Azobisisobutyronitrile (AIBN, AR), propylene glycol methyl ether (PGM, AR), tetrahydrofuran (THF, AR), methyl methacrylate (MMA, AR), ethyl acrylate (EA, AR), acrylic acid (AA, AR) and KBr (GR) were purchased from Tianjin Guangfu Technology Development Co., Ltd.. Absolute ethanol was purchased from Tianjin Fuyu Fine Chemical Co., Ltd.. Xylene (AR) was purchased from Tianjin Damao Chemical Reagent Factory. Peptone (BR) was purchased from Beijing Aoboxing Biotechnology Co., Ltd.. NaCl (AR) was purchased from Tianjin Tianda Chemical Reagent Factory. *Escherichia coli* and *Staphylococcus aureus* were purchased from Shanghai Preservation Biotechnology Center. Marine bacteria were extracted from the sea area of Dalian City, Liaoning Province. *Halamphora* sp. and *Nitzschia Closterium* were purchased from Xiamen University. Deionized water was homemade in the laboratory. All reagents are used directly without treatment.

### 2.2. Synthesis of PBAB polymer

The specific synthetic route of the PBAB polymer that was synthesized by a two-step method is shown in Fig. 1a. First, MMA, EA, IBOMA, and AA were used to prepare isobornyl methacrylate polymer (PBA<sub>COOH</sub>) via free radical polymerization, which was initiated by AIBN. Subsequently, PTPB and PBA<sub>COOH</sub> were used to prepare PBABs. The preparation methods of other polymers remain the same, and only the monomer changes.

#### 2.2.1. Synthesis of the PBA<sub>COOH</sub> polymer

PBA<sub>COOH</sub> was synthesized by free radical polymerization. First, xylene (60 g, 0.565 mol) and PGM (60 g, 0.665 mol) were added into a four-necked flask equipped with a condenser, stirrer, thermometer, and dropping funnel, and then the mixture was heated to 95 °C. Subsequently, MMA (30.75 g, 0.307 mol), EA (30.75 g, 0.307 mol), IBOMA (20 g, 0.089 mol), AA (3.5 g, 0.0485 mol), and AIBN (2 g, 0.012 mol) were thoroughly mixed and then added dropwise to the flask. After that, xylene (15 g, 0.141 mol) and BPO (0.5 g, 0.002 mol) were added to the flask within 30 min, and then stirring was continued for 2 h to obtain PBAs. Finally, excess ice ethanol was added to the PBAs to precipitate the polymers, and the polymers were redissolved in the solvent to adjust the solid content within a proper range.

#### 2.2.2. Synthesis of the PBAB polymer

PBABs were prepared under the following reaction conditions: PTPB (15 g, 0.0467 mol), xylene (60 g, 0.565 mol), and THF (40 g, 0.555 mol) were added to a flask equipped. The temperature of the mixture was raised to 100 °C, PBA polymers with different AA contents were added and the temperature was kept constant. The mixture was continuously stirred for 12 h, and after approximately 2 h, the mixture changed from a milky white suspension to a brown transparent system. After the reaction, the ungrafted PTPB was removed by filtration, and then the solid content of the system was adjusted to an appropriate range by vacuum distillation. Table 1 summarizes the content of each component of different PBAB polymers.

## 3. Results and discussion

### 3.1. Structure and characterization of PBABs

The specific synthetic route of the PBAB polymer that was synthesized by a two-step method is shown in Fig. 1a. To determine the composition and molecular structure of the PBAB polymer, FT-IR and <sup>1</sup>H NMR were performed. The infrared spectra of PBAs, PBA<sub>COOH</sub> and PBABs are shown in Fig. 1b. For the three polymers,

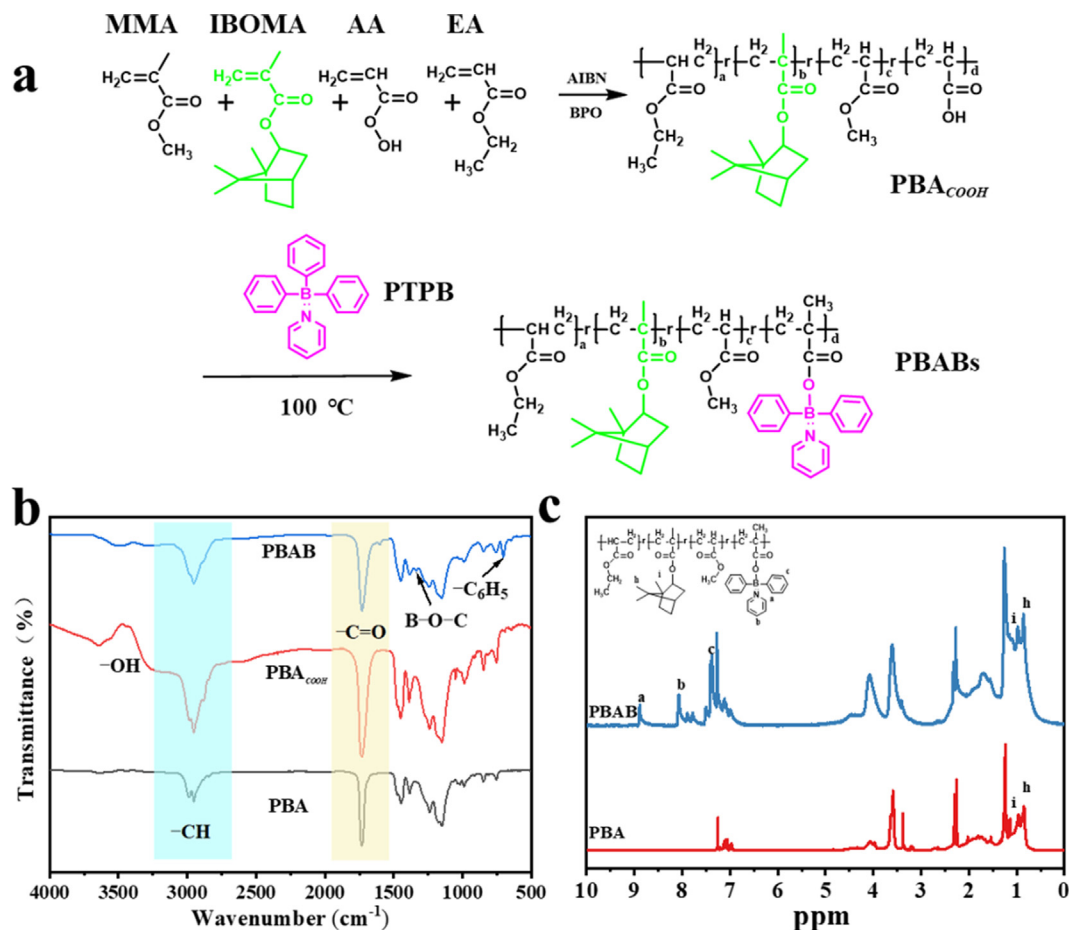


Fig. 1. (a) Synthetic route of PBABs. (b) FT-IR spectra of PBAs, PBA<sub>COOH</sub> and PBABs. (c) <sup>1</sup>H NMR spectra of PBAs and PBABs.

Table 1

Formula of different copolymers, Mw, PDI and solid content.

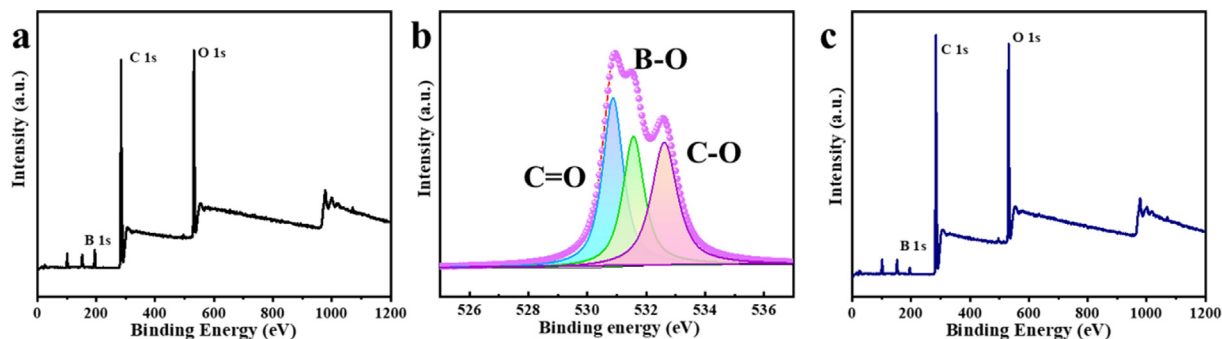
| Samples | MMA   | EA    | IBOMA | AA  | PTPB | Mw     | PDI  | Solid content/% |
|---------|-------|-------|-------|-----|------|--------|------|-----------------|
| PAC     | 50    | 50    | 0     | 0   | 0    | 26,948 | 1.83 | 46.45           |
| PBA     | 40    | 40    | 20    | 0   | 0    | 30,404 | 1.83 | 46.53           |
| PBAB-1  | 36.75 | 36.75 | 20    | 1.5 | 5    | 32,806 | 1.74 | 47.37           |
| PBAB-2  | 33.75 | 33.75 | 20    | 2.5 | 10   | 29,499 | 1.72 | 46.17           |
| PBAB-3  | 30.75 | 30.75 | 20    | 3.5 | 15   | 32,475 | 1.77 | 47.42           |
| PACB    | 40.75 | 40.75 | 0     | 3.5 | 15   | 31,501 | 1.81 | 47.73           |

the peak at 2950 cm<sup>-1</sup> represents -CH bonds, while the peak at 1740 cm<sup>-1</sup> corresponds to the carbonyl group on the ester bond. For PBA<sub>COOH</sub>, the characteristic peak in the range of 3200 ~ 3400 cm<sup>-1</sup> is ascribed to the stretching vibration of the -OH groups in the carboxyl moieties. By comparing the spectra of PBAs, PBA<sub>COOH</sub> and PBABs, a new peak is observed at 1335 cm<sup>-1</sup>, which is due to the presence of the B-O-C bond. Furthermore, it can be significantly noted that the hydroxyl peak in the PBA<sub>COOH</sub> spectrum does not appear in the PBAB spectrum, indicating the reaction between the carboxyl group in the PBA polymer and PTPB. The FT-IR results show that the PBAB polymer was successfully synthesized.

Furthermore, the <sup>1</sup>H NMR spectra are presented in Fig. 1c. For the PBAs, the peaks at  $\delta = 0.95$  and  $\delta = 0.85$  represent the borneol group on the side chain of the polymer. For the PBABs, there are three new peaks in the <sup>1</sup>H NMR spectrum of PBAs located at  $\delta = 8.9$ ,  $\delta = 8.0$ , and  $\delta = 7.2$ , which all correspond to the grafted PTPB groups on the PBAB polymer. The <sup>1</sup>H NMR and FT-IR results con-

firm the successful synthesis of PBAB polymers. The molecular weight (Mw) and the polydispersity (PDI) of the synthesized polymers were measured by GPC, as shown in Table 1. The molecular weights (Mw) of the polymers are distributed between  $2.6 \times 10^4$  and  $3.2 \times 10^4$ , and the polydispersity (PDI) is concentrated between 1.72 and 1.83.

The XPS survey spectrum of the PBAB coating in Fig. 2a shows that the coating contains C, O, and B elements. The high-resolution O 1s spectra of the PBAB coating before immersion in artificial seawater were checked, and are presented in Fig. 2b. The PBAB coating exhibits O species at 530.8 eV and 532.7 eV, which are attributed to C=O and C-O, respectively. In addition, a new signal located at 531.6 eV was assigned to B-O, which strongly supports the successful preparation of PBAB coatings. However, the characteristic peak of B 1s on the surface of the PBAB coating immersed in artificial seawater for 14 days decreased rapidly (Fig. 2c), indicating that the side chains were hydrolyzed and fell off, which suggested that the coating surface had been hydrolyzed.



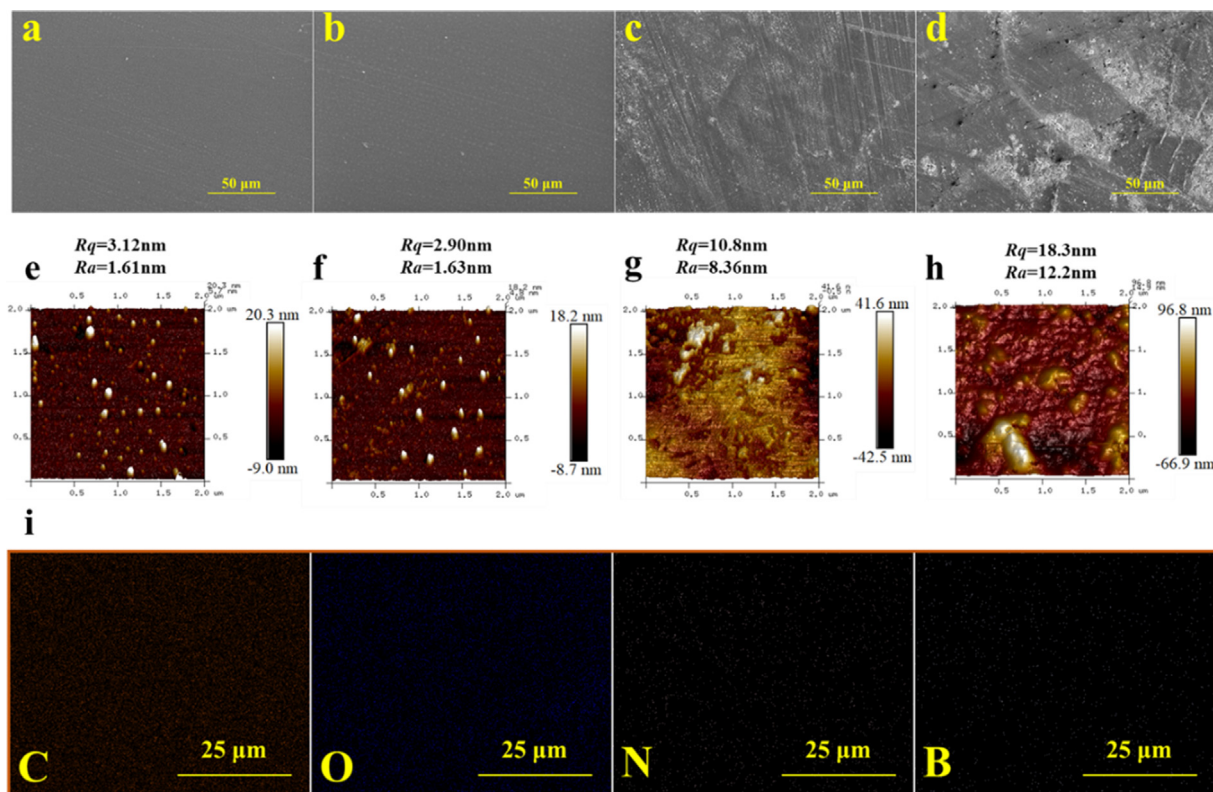
**Fig. 2.** (a) XPS wide-scan spectrum of the PBABs. (b) high-resolution O 1 s spectra. (c) XPS wide-scan spectrum of PBABs immersed in artificial seawater for 14 days.

### 3.2. Morphology of the PBAB coatings

It is undeniable that the microstructure and roughness of the coating surface have an important influence on the adhesion of fouling organisms. Therefore, the surface morphology of the PBAB coating before and after immersion in artificial seawater was observed by SEM and AFM.

SEM images of PBAB and PACB coatings before and after immersion in artificial seawater for 14 days are shown in Fig. 3 a–d. The surfaces of PACBs and PBABs are relatively smooth and flat without wrinkles before immersion, which indicates that both coatings have excellent film-forming properties. Simultaneously, the two coatings have hairy protrusions on the surface, which can be attributed to the hydrolyzable diphenyl borane pyridine side group on the polymer side chain. The two coatings have hairy protrusions on the surface, which can be attributed to the hydrolyzable diphenyl borane pyridine side group on the polymer side chain. After

being immersed in artificial sea water for 14 days, the protrusions on the coating surface disappeared, indicating that the hydrolyzable side chains were hydrolyzed, so the protrusions disappeared [33]. Surprisingly, the surface morphology of the PACB coating without IBOMA monomer and the PBAB coating immersed in ASW for 14 days was completely different. There are many fluctuations and protrusions on the surface of the PACB coating, which may be attributed to the fast and uneven hydrolysis of the hydrolyzable diphenyl borane pyridine side group on the surface. However, the PBA coating remained relatively flat after immersion for 14 days, although the surface became slightly rough. This is due to the hydrophobicity of the IBOMA monomer contained in the polymer, which can delay the hydrolysis rate of the coating, thus, maintaining a uniform hydrolysis rate and a relatively flat immersion time. Fig. 3i shows the mapping picture of the PBAB coating. C, O, N, and B are evenly distributed on the surface of the coating without local aggregation.



**Fig. 3.** SEM images of the (a) PBAB-3, and (c) PBAB-3 coatings after immersion in artificial seawater for 14 days, and (b) PACB and (d) PACB coatings after immersion in artificial seawater for 14 days. (e and f) Atomic force microscopy (AFM) images of the PBAB-3, and PACB coatings and (g and h) PBAB-3, and PACB coatings after immersion in artificial seawater for 14 days.

Similarly, the immersed PBAB coating has a smoother surface than the PACB coating, which is proven by the AFM test (Fig. 3e–h). Specifically, the values of  $R_a$  and  $R_q$  of the two coatings changed before and after hydrolysis. However, the changes in the  $R_a$  and  $R_q$  values of the PBAB coating are significantly lower than those of the PACB. Compared with the PACB coating, the PBAB coating surface is relatively flat and not too convex. The AFM characterization results are consistent with the SEM results, indicating that the PBAB coating can maintain better surface morphology during hydrolysis, which helps to resist fouling organisms adhesion.

### 3.3. Antifouling tests of the PBAB coatings

Marine bacteria, diatoms, and mussels were selected as adhesion models to evaluate the antifouling performance of coatings. Compared with the control sample PAC, the PBA and PBAB coatings have a strong inhibitory effect on marine bacterial adhesion (Figure S1). However, the PACB coating without IBOMA monomer showed poor adhesion of marine hybrid bacteria, nearly the same as the control sample PAC.

To further evaluate the broad-spectrum antibacterial properties of the coating, *E. coli* (gram-negative bacteria) and *S. aureus* (gram-positive bacteria) were selected as representatives of the broad-spectrum antibacterial evaluation (Fig. 4a). The results showed that the PBAB coating has significant adhesion inhibition properties for *E. coli* and *S. aureus*. By calculating the number of bacterial communities on the plate, the inhibition rates of PBAB-1, PBAB-2, and PBAB-3 against *E. coli* are 98.1%, 97.5%, and 95.1%, respectively, while the inhibition of the PACB is only 7.2% (Figure S2). Comparably, the inhibitory rates of PBAB-1, PBAB-2, and PBAB-3 against *S. aureus* were slightly lower, at 91.7%, 88.7%, and 81.1%, respectively; however, they were still much higher than those of PACBs. The results indicate that the IBOMA component contained in the PBAB coating can effectively inhibit the adhesion of bacteria, which causes major marine biofouling. The effect of the coating on bacterial growth is shown in Figure S3. For *E. coli* (Figure S3a), the bacterial growth concentrations of each component were almost the same within the first 6 h. After that, the PBAB coating containing the IBOMA component had a significant inhibitory effect on bacterial growth, and the IBOMA component released into the bacterial solution after coating hydrolysis was the main reason for this result. *S. aureus* had the same results (Figure S3b). The experimen-

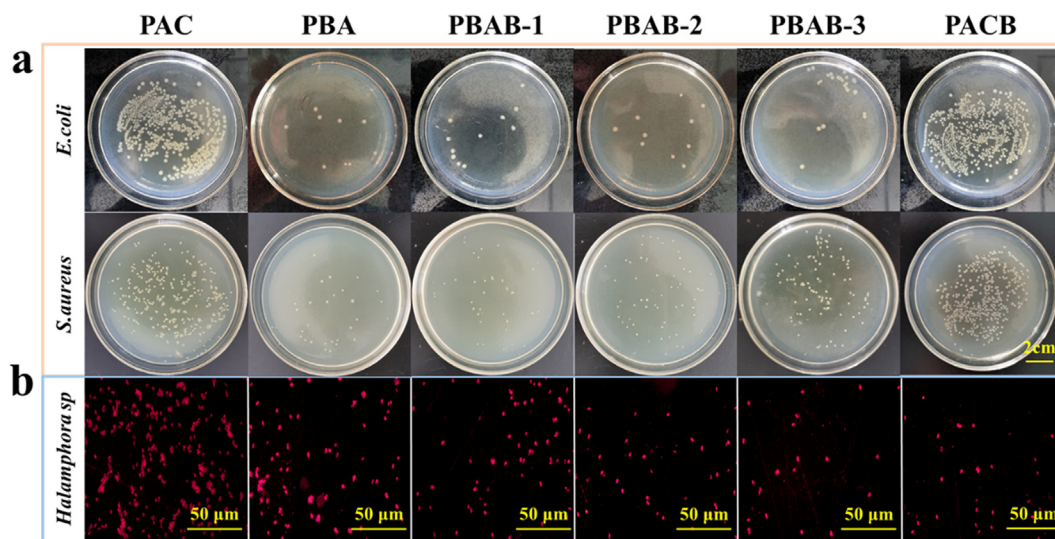
tal bacterial concentration results were consistent with the antibacterial results, indicating that PBAB coating had good antibacterial properties.

In addition, benthic diatoms *Halamphora sp* and *Nitzschia closterium* were used as typical diatom models to evaluate the anti-diatom adhesion resistance of the coating. Fig. 4b shows that the number of diatoms attached to the surface of the PBA and PBAB coatings after 24 h was significantly less than that of the control PAC sample. This result can be attributed to the presence of the IBOMA component that makes the formation of the primary fouling biofilm more difficult, significantly reducing the adhesion of bacteria and ultimately affecting the adhesion of diatoms on the surface. For PBAs and PBABs, with the increase in PTPB components, the number of diatoms was significantly reduced, indicating that the resistance of the material to diatom adhesion was gradually improved. The performance of PACB coatings without IBOMA components is also excellent, but the fast hydrolysis rate and poor ability to maintain a smooth surface of PACBs do not propose long-term antifouling use. The diatom resistance of different coatings is shown in Figure S4. PBAB-3 is the best of all coatings, reaching 92.76%.

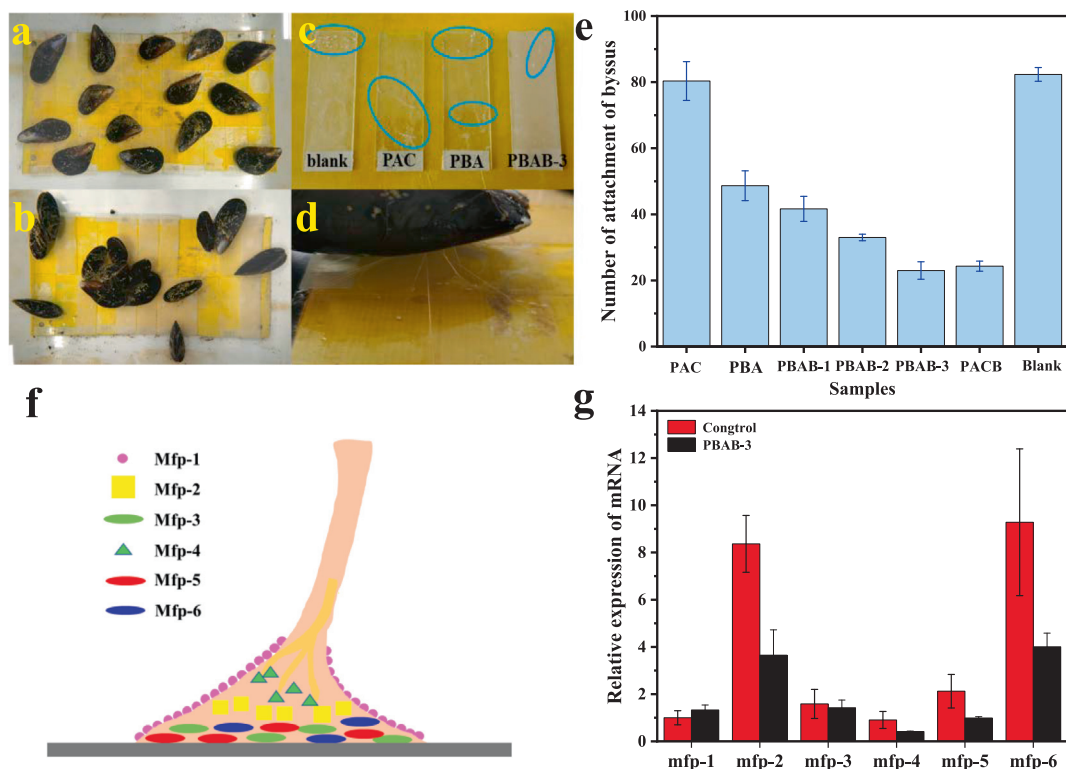
Moreover, fluorescence images of the adhesion of *Halamphora sp* on all coating surfaces showed strong adhesion and obvious agglomeration, which is consistent with the habit of *Halamphora sp*. PBABs still have excellent antifouling performance after 14 days (Figure S5). For further investigation, another diatom *Nitzschia closterium* was selected for a 14-day adhesion test, and the result was consistent with *Halamphora sp*, indicating that the PBAB coating has potential for long-term application.

Large-scale biofouling, especially hard-shelled organisms such as mussels, is extremely harmful to ship navigation. Therefore, adult mussels captured from the Dalian Sea were selected to test the antifouling performance of the coating.

The mussels were placed on a random sample plate for 24 h to observe their movements, and the results are shown in Fig. 5a–b. The mussels changed their position and moved considerably after 24 h, which was attributed to the selective adhesion behavior of the mussels. From the adhesion results, it can be noted that the mussels prefer to adhere to the blank glass plate and the PAC coating. Fig. 5c–d are digital photos of the attached mussels on the glass sheet, and Fig. 5e is the average number of byssuses attached to each coating after 3 experiments.



**Fig. 4.** (a) Antibacterial image of different PBAB and PACB coatings for *E. coli* and *S. aureus*. (b) Fluorescence microscopy images of different coatings immersed in *Halamphora sp* algae solution for 24 h.



**Fig. 5.** (a) Image of mussels randomly scattered on the surface of the samples. (b) Image of mussels attached to the surface of the sample after 24 h. (c) The byssus plates on the blank, PAC, PBAB-3, and PACB coatings. (d) Mussels attached to glass sheets. (e) The average number of byssus plates on different samples in 3 experiments. (f) Localization of adhesive proteins in the byssus. (g) The relative expression level of adhesion protein-related RNA in glass and PBAB-3 coatings.

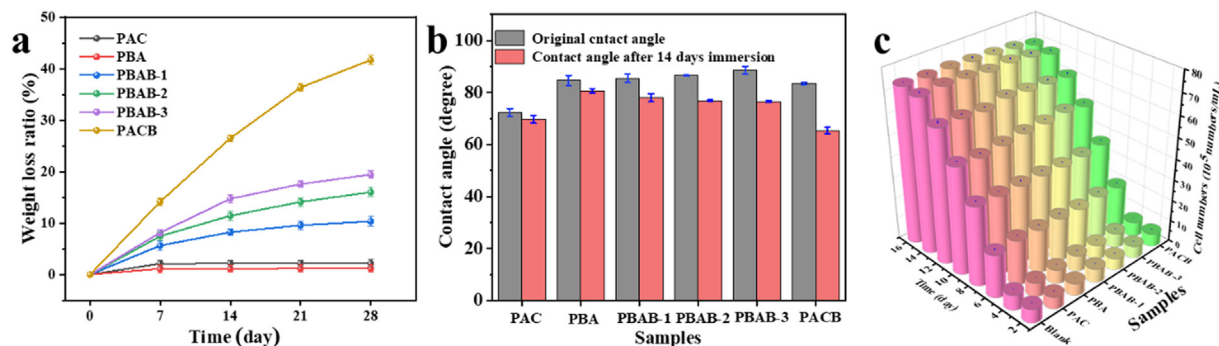
Multiple different adhesion proteins have been identified in mussel silk discs. Here, the six adhesion proteins: mfp-1, mfp-2, mfp-3, mfp-4, mfp-5, and mfp-6 were quantitatively analyzed. Previous research results indicate that mfp-1 exists on the surface of the silk fiber and plays a protective role, while mfp-2 is only distributed in the silk disc to form adhesion spots. The three proteins mfp-3, 5, and 6 are mainly distributed on the contact surface between the byssus and the substrate, and these 3 proteins play a leading role in adhesion to the substrate (Fig. 5f)[34]. The protein quantitative analysis results are shown in Fig. 5g. Surprisingly, the expression of mfp-3, mfp-5, and mfp-6 proteins in mussels placed on PBAB coatings was significantly lower than that in control samples, indicating that PBAB coatings can inhibit the expression of mussel adhesion proteins and consequently resist mussel adhesion, which is consistent with the results of mussel adhesion experiments.

### 3.4. Surface interface and environmentally friendly characteristics

The hydrolysis rate and surface wettability have an important influence on the overall performance of the self-polishing antifouling coating since hydrolysis rates that are too fast or too slow are not conducive to the antifouling material. The hydrolysis rate of the coating was evaluated by the mass loss of coating immersion, and results are shown in Fig. 6a. The hydrolysis rate of PAC and PBA coatings without the PTPB component is very low. With increasing of PTPB component content, the hydrolysis rate of the coating also increased, and the overall hydrolysis rate was relatively gentle. Moreover, the hydrolysis rate of PACBs is much higher than that of PBABs. Specifically, the weight loss in artificial seawater within 24 days reaches 42%, while that of PBAB-3 is only 19%. It is well known that hydrophobic coatings have less contact with water, which could lead to a low weight loss ratio, and vice

versa. The reason is that IBOMA is a hydrophobic molecule that can significantly reduce the hydrolysis rate of the coating, ensuring that the coating maintains long-term antifouling performance. The coating thickness changes before and after immersion are shown in Figure S6. After 7 days of artificial seawater immersion, the thickness of the self-polishing PBAB coating decreased significantly, which was attributed to the hydrolysis of the side chains of PBABs. In addition, the contact angles before and after coating immersion are shown in Fig. 6b, indicating that the original hydrophilic PAC became hydrophobic when the IBOMA molecule was introduced into the PAC coating. The contact angles of PBABs and PACBs are significantly reduced after immersion. The reason for this phenomenon is that the side group of diphenylpyridine reacted with sodium in weakly alkaline seawater and then hydrolyzed in artificial seawater. However, the contact angle of the PBAB coating changed much less before and after immersion, due to the hydrophobicity of the IBOMA molecule. Therefore, the PBAB coating can maintain a relatively stable hydrophobic surface.

To evaluate the toxic effect of hydrolysate on marine organisms, the environmentally friendly performance of the coating was investigated by the growth characteristics of diatoms and bacteria. As shown in Figure S7, all coatings and blanks were soaked in *Nitzschia closterium* solutions for 16 days. After incubation for 16 days, no significant difference in the concentration of *Nitzschia closterium* could be detected on the PBAB coatings and blank. Digitization, the cell numbers of seaweed containing blank, PACs, PBABs, PBAB-1, PBAB-2, PBAB-3, and PACB coatings were  $73.57 \times 10^5$ ,  $73.27 \times 10^5$ ,  $73.33 \times 10^5$ ,  $72.73 \times 10^5$ ,  $72.10 \times 10^5$ ,  $71.27 \times 10^5$ , and  $69.27 \times 10^5$  numbers/mL, respectively (Fig. 6c). The same results were obtained for the number of colonies, indicating that the coatings are nontoxic to biofouling and environmentally friendly. Adhesion is of great significance to marine antifouling coatings. We conducted adhesion tests on different



**Fig. 6.** (a) Weight loss ratios of PAC, PBAB, and PACB coatings. (b) The contact angle of different coatings before and after immersion in artificial seawater for 14 days. (c) Diatom growth numbers in solutions containing different coatings.

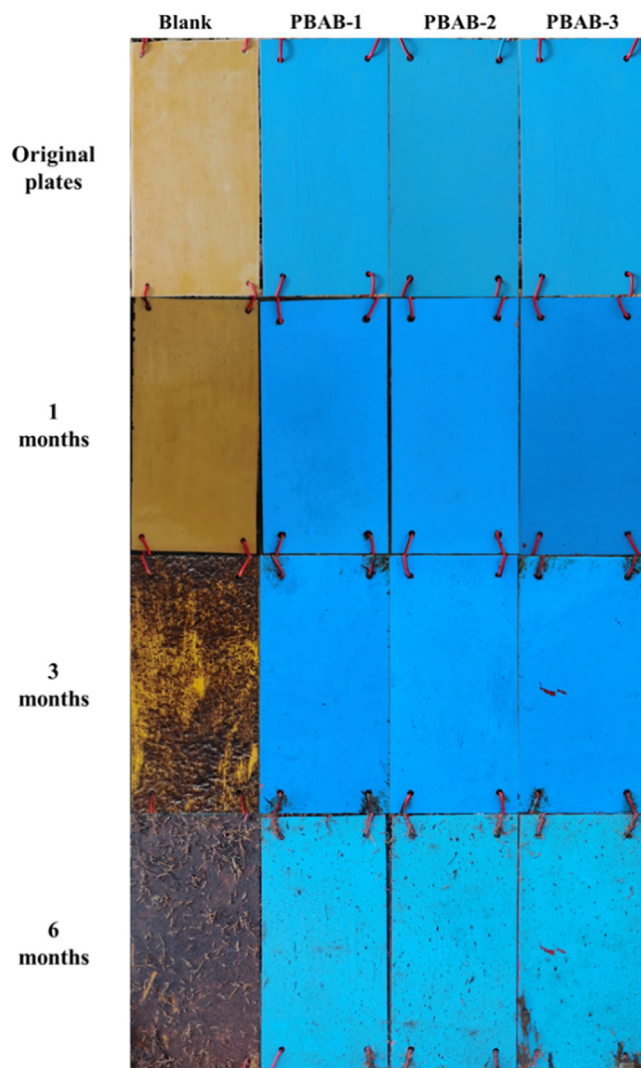
coatings. The results are shown in Figure S8. After adding PTPB, the adhesion of the coating decreased slightly, but the overall adhesion remained above 4.5 MPa.

### 3.5. Antifouling paints raft test of different PBABs

To explore the antifouling performance of PBAB coatings in an actual marine environment, marine antifouling experiments were carried out in the Yellow Sea of China (photographs of the coated and blank plates are shown in Fig. 7). The yellow plate is the untreated blank as a control group, and the rest are different samples (PBAB-1, PBAB-2, PBAB-3). Experiments were carried out from February to July when marine organisms flourished. In the first month, none of the plates had obvious attachment of biofouling. However, the plates in the blank group were black, owing to attaching marine bacteria to its surface, forming a layer of microbial membrane, which forms the basis to attach the algae and other large biofouling. In contrast, the PBAB coating surface is clean because IBOMA can effectively block the attachment of bacteria. The comparison is particularly significant after three months. The control group was full of biofouling, while the surface of PBABs remained smooth and only had some attached algae. Six months later, the control sample surface was covered by marine organisms, while the PBAB coating surface in addition to slight fouling, had no obvious fouling organisms attached. The excellent antifouling performance of PBABs is attributed to the synergistic antifouling effect of IBOMA and PTPB. IBOMA can effectively resist the formation of primary biofouling, while the PTPB group of the copolymer side chain hydrolysis can effectively remove fouling organisms attached to the surface. At the same time, we found an interesting phenomenon. In the third month of taking pictures and recording, we found trace damage on the surface of the PBAB-3 coating (probably due to bumps). In the sixth month of inspection, the damage did not expand. This accidental discovery indicates that the local damage of the PBAB coating will not affect the performance of the overall coating, which is of great significance for practical applications.

## 4. Conclusions

Self - renewal coating PBABs with antibacterial and antifouling properties were successfully synthesized by a two-step method. The PBAB polymer has good film-forming properties and a stable hydrolysis rate in seawater, which ensures that it can stably form a film in seawater and is gradually polished to resist the adhesion of fouling organisms. The IBOMA component contained in the PBAB polymer can significantly reduce the hydrolysis rate of the coating to have a longer service life and simultaneously, provide excellent antibacterial properties. Antibacterial and anti-diatom tests proved



**Fig. 7.** Photographs of the PBAB coatings and the blank immersed in the Yellow Sea of China for 6 months.

that PBABs can significantly inhibit the formation of primary marine fouling biofilms. The results of adhesion experiments and protein quantitative analysis of mussels showed that the coating can effectively inhibit the adhesion of large fouling organisms. Based on the excellent antifouling ability of the PBAB coating, the antifouling mechanism is inferred as follows: First, the presence of an IBOMA component can significantly inhibit the formation of

the bacterial membrane so that small fouling organisms (diatoms, etc.) have no suitable attachment environment. Second, large biofouling cannot obtain food sources on the surface of PBAB coatings to reach their unwillingness to attach. Finally, the biofouling that has adhered to the surface of the PBAB coatings can be removed by hydrolysis to renew the surface[33]. Marine experiment proved the excellent antifouling performance of PBAB coatings in seawater. In addition, there was no significant impact on the growth of diatoms on the PBAB material immersed in the diatom solution, which reveals the environmentally friendly characteristics of this material. In summary, PBAB coatings are a kind of antifouling coating with applicable potential.

#### CRedit authorship contribution statement

**Fan Song:** Conceptualization, Methodology, Investigation, Software, Writing – original draft. **Jun Wang:** Supervision, Funding acquisition. **Linlin Zhang:** Conceptualization, Methodology, Investigation, Software, Writing – original draft. **Rongrong Chen:** Supervision, Writing – review & editing. **Qi Liu:** Supervision, Writing – review & editing. **Jingyuan Liu:** Data curation, Resources. **Jing Yu:** Data curation, Resources. **Peili Liu:** Project administration. **Jizhou Duan:** Project administration.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2021.12.187>.

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